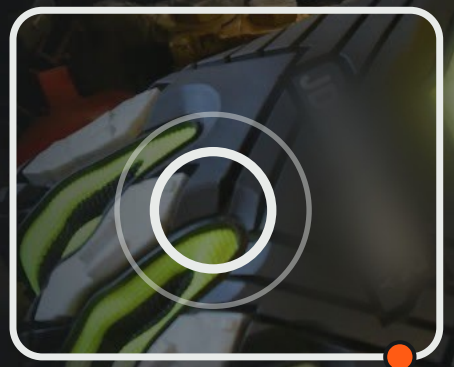
praxis

STEP 4/9

Instructions that can finally see how you're doing

An AI coach for new maintenance technicians. It watches the work through a first-person camera and says the next step.



Seat the clamp past the hose bead

STARTUPS @ HARVARD · 20 OCT 2026 · MONISH SARAVANA

CONCEPT MOCK · STOCK PHOTO

THE PROBLEM

A new maintenance technician needs about a year of a senior tech's time to work alone

9–14

months before a new technician pays for themselves, in a survey of 35 repair stations.

$10–15k

to hire and ramp up one maintenance technician, counting onboarding and lost productivity.

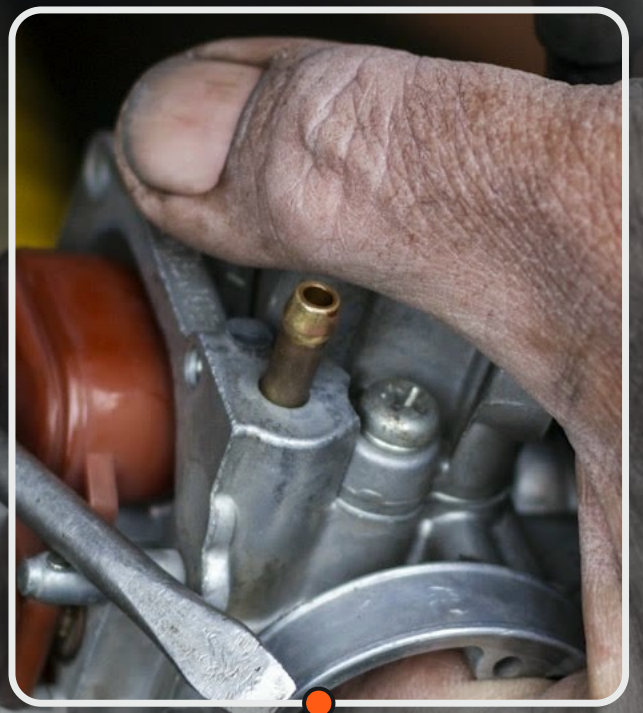
~255,000 openings a year for US maintenance and repair techs, and every new hire pulls a senior tech off the floor.

ARSA member survey of 35 repair stations, 2017 · Reliability magazine (A. Gager) · BLS Occupational Outlook Handbook

THE PRODUCT

Praxis watches the work and coaches one step at a time

LOAD	the manuals and checklists a shop already has
COACH	on a phone or glasses, one short line at a time
VERIFY	critical steps and keep the record



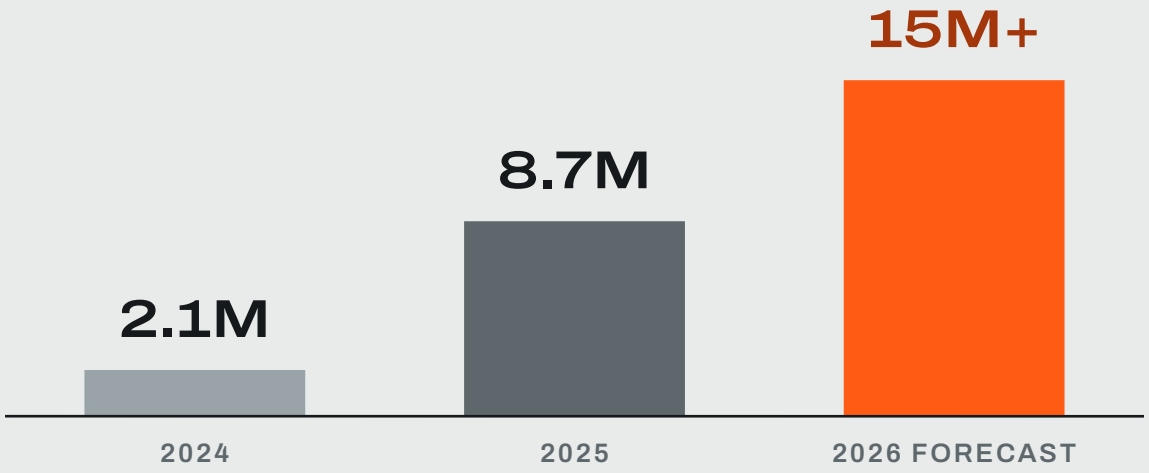
Loosen the four float-bowl screws a half turn each

CONCEPT MOCK · STOCK PHOTO

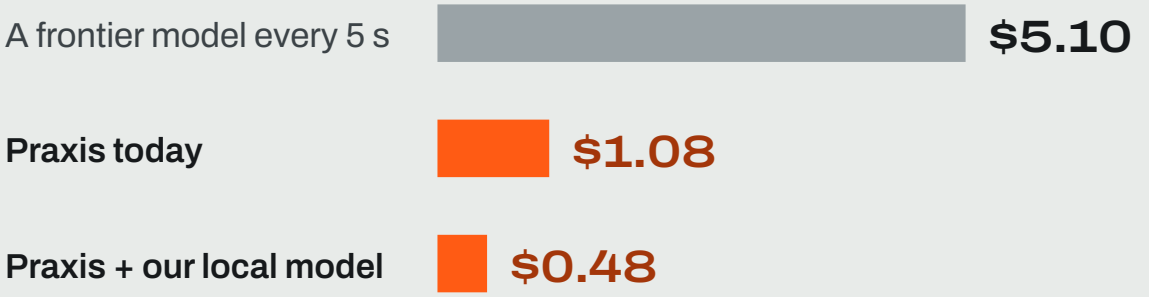
WHY NOW

2026: cameras moved onto faces, and watching work got cheap

CAMERAS ON FACES · AI GLASSES SHIPPED A YEAR



COST TO WATCH ONE HOUR OF WORK



and the price of the same AI keeps falling about 10× a year

MAY 2026

Autodesk buys MaintainX, a maintenance app, for $3.6B

SEPT 2026

Meta opens its glasses' camera to outside apps

NOV 2026

Android XR glasses reported to ship

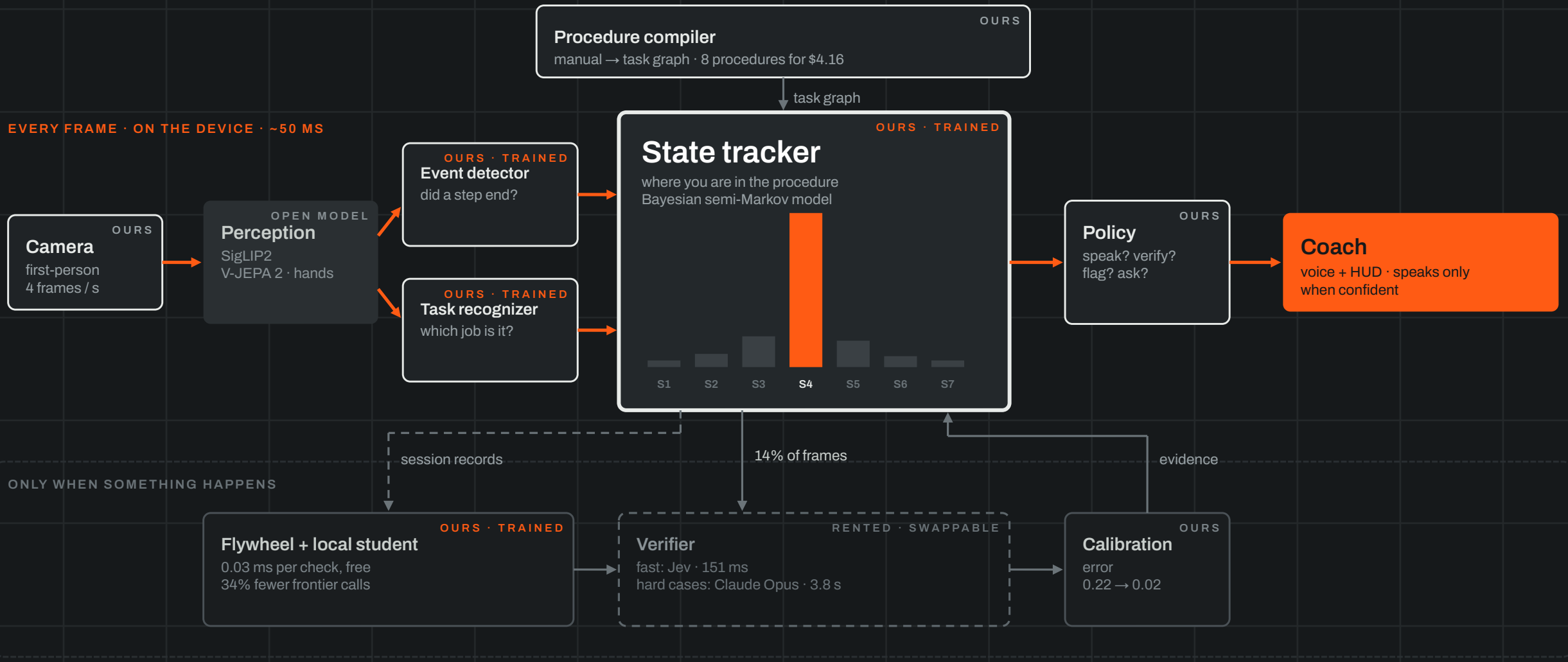
Q1 2027

Viture + NVIDIA coaching glasses ship

Omdia (2025 shipments, +322%; 2026 forecast) · Praxis run logs · a16z, LLMflation · Autodesk, Meta, 9to5Google, Viture releases

HOW IT WORKS · ENGINE v0

The frontier model is one swappable part. The tracking engine around it is ours.

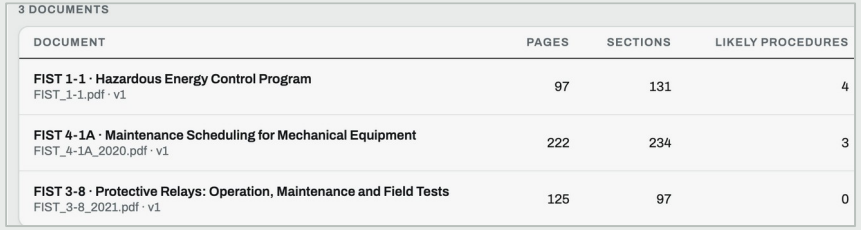


Ours · Open model, on device · Rented, swappable

OURS · TRAINED: gets better with more labeled recordings

48 held-out clips · one run each

A shop uploads the manuals it already has. Praxis turns each procedure into a coach



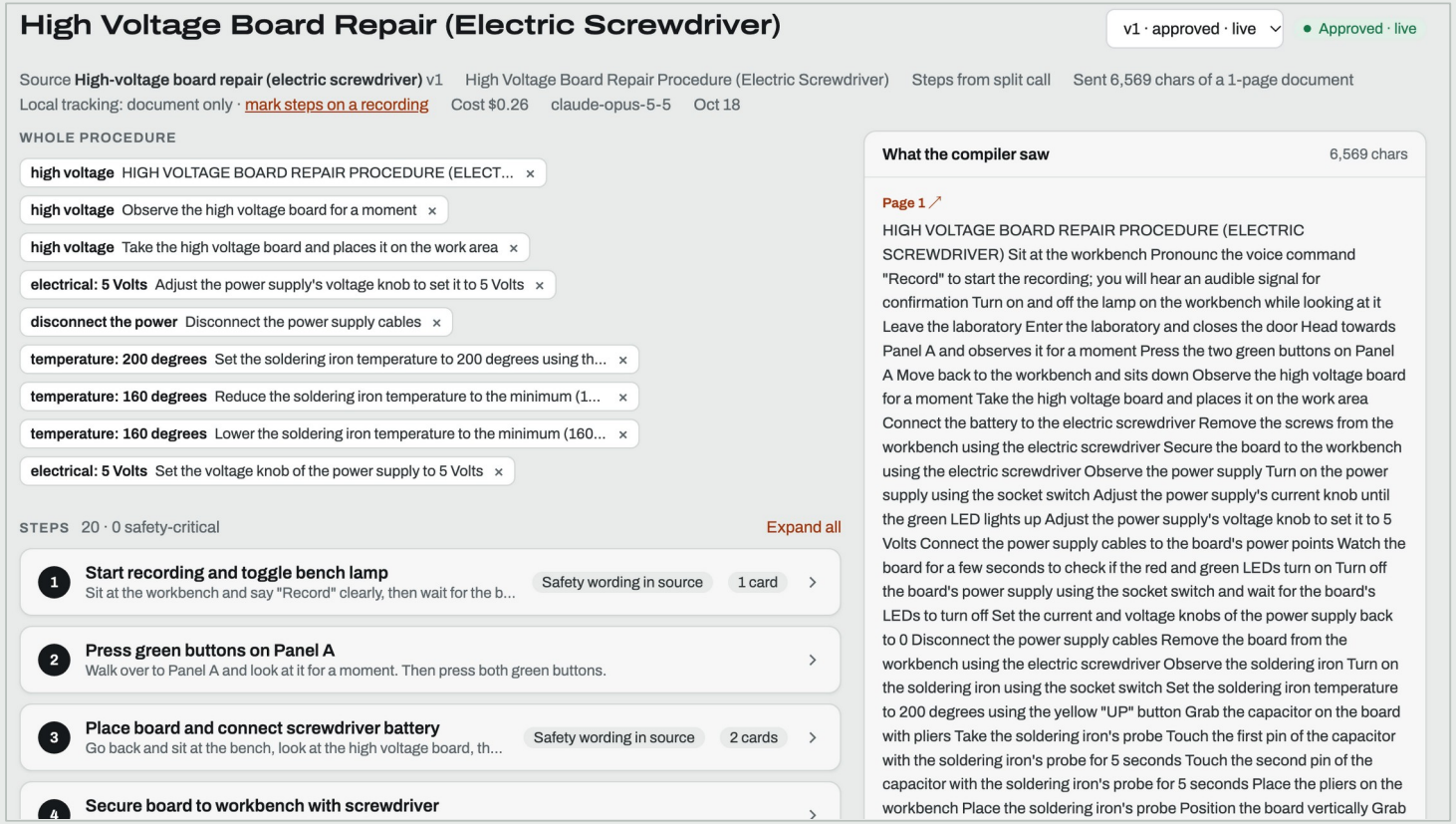
3 DOCUMENTS

DOCUMENT	PAGES	SECTIONS	LIKELY PROCEDURES
FIST 1-1 · Hazardous Energy Control Program FIST_1-1.pdf · v1	97	131	4
FIST 4-1A · Maintenance Scheduling for Mechanical Equipment FIST_4-1A_2020.pdf · v1	222	234	3
FIST 3-8 · Protective Relays: Operation, Maintenance and Field Tests FIST_3-8_2021.pdf · v1	125	97	0

1. **Upload the manuals**
   A 97-page safety manual is indexed in 0.34 s on a laptop, for $0. Its procedures are found automatically.

2. **Review the coached steps**
   Each procedure compiles for about $0.19, with cards quoted from the manual. A manager approves it first.

3. **Coach on a phone or glasses**
   Live tracking, voice control, Meta and Viture glasses connections, screen or voice-only modes.



High Voltage Board Repair (Electric Screwdriver)

v1 · approved · live · Approved · live

Source **High-voltage board repair (electric screwdriver)** v1 · High Voltage Board Repair Procedure (Electric Screwdriver) · Steps from split call · Sent 6,569 chars of a 1-page document
Local tracking: document only · mark steps on a recording · Cost $0.26 · claude-opus-5-5 · Oct 18

WHOLE PROCEDURE

- **high voltage** HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECT...
- **high voltage** Observe the high voltage board for a moment
- **high voltage** Take the high voltage board and places it on the work area
- **electrical: 5 Volts** Adjust the power supply's voltage knob to set it to 5 Volts
- **disconnect the power** Disconnect the power supply cables
- **temperature: 200 degrees** Set the soldering iron temperature to 200 degrees using th...
- **temperature: 160 degrees** Reduce the soldering iron temperature to the minimum (1...
- **temperature: 160 degrees** Lower the soldering iron temperature to the minimum (160...
- **electrical: 5 Volts** Set the voltage knob of the power supply to 5 Volts

STEPS 20 · 0 safety-critical — Expand all

1. **Start recording and toggle bench lamp** — Sit at the workbench and say "Record" clearly, then wait for the b... — Safety wording in source — 1 card
2. **Press green buttons on Panel A** — Walk over to Panel A and look at it for a moment. Then press both green buttons.
3. **Place board and connect screwdriver battery** — Go back and sit at the bench, look at the high voltage board, th... — Safety wording in source — 2 cards
4. **Secure board to workbench with screwdriver**

What the compiler saw — 6,569 chars

Page 1

HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECTRIC SCREWDRIVER) Sit at the workbench Pronounc the voice command "Record" to start the recording; you will hear an audible signal for confirmation Turn on and off the lamp on the workbench while looking at it Leave the laboratory Enter the laboratory and closes the door Head towards Panel A and observes it for a moment Press the two green buttons on Panel A Move back to the workbench and sits down Observe the high voltage board for a moment Take the high voltage board and places it on the work area Connect the battery to the electric screwdriver Remove the screws from the workbench using the electric screwdriver Secure the board to the workbench using the electric screwdriver Observe the power supply Turn on the power supply using the socket switch Adjust the power supply's current knob until the green LED lights up Adjust the power supply's voltage knob to set it to 5 Volts Connect the power supply cables to the board's power points Watch the board for a few seconds to check if the red and green LEDs turn on Turn off the board's power supply using the socket switch and wait for the board's LEDs to turn off Set the current and voltage knobs of the power supply back to 0 Disconnect the power supply cables Remove the board from the workbench using the electric screwdriver Observe the soldering iron Turn on the soldering iron using the socket switch Set the soldering iron temperature to 200 degrees using the yellow "UP" button Grab the capacitor on the board with pliers Take the soldering iron's probe Touch the first pin of the capacitor with the soldering iron's probe for 5 seconds Touch the second pin of the capacitor with the soldering iron's probe for 5 seconds Place the pliers on the workbench Place the soldering iron's probe Position the board vertically Grab

Screenshots of the Praxis company backend · Hydro Plant demo: public Reclamation FIST manuals · ENIGMA-51 repair procedures

HOW FAR ALONG · MEASURED

It already beats a frontier model on its own. The one bottleneck is training data

54% vs 48%

step accuracy against the same frontier model called every 5 seconds

1/5 the cost

$0.018 vs $0.085 per minute of video, because it only asks a model when something happens

11 of 15

technician questions on an industrial repair job answered correctly from the company's own manuals

0

times confidently wrong when the view was blurred or blocked. It asks for a better view instead

THE BOTTLENECK: DATA

Our step models learned from only a handful of videos per procedure.

More videos, better models: 31% → 51% on videos they had never seen, and still rising.

Catching mistakes needs real maintenance examples to learn from.

Buying labeled data and GPU time to retrain is most of the $10k.

48 held-out clips of public first-person procedure video · one run per system · questions: ENIGMA-51 industrial repair, blind-judged

COMPETITORS

Rolling out a new procedure: what it takes today, and what it takes with Praxis

Procedure apps MaintainX, Augmentir	Type the steps into the app	The tech reads them on a phone	Ticks a box when done	Nobody sees whether the step was done right
Fixed cameras Retrocausal, Invisible AI	Mount a camera at one workstation	Record and label runs to train it	It watches that one station	Can't follow a tech around a machine
First-person AI glasses Strivr, Viture + NVIDIA	Their team films your procedures	They train a model for your site	Coach through their glasses	Filming and training for every customer
Praxis	Upload the manual you already have	Approve the coached steps, about $0.19 each	Coach on a phone or glasses	**Verified and recorded, with no filming**

Vendor product pages and docs, checked 17 Oct 2026 · Retrocausal: record, label, train · Strivr: per-customer capture and model · detail in appendix A7

WHY NOBODY OWNS THIS NICHE

Meta, Google and Apple won't specialize, so we will. Their glasses become our hardware

15M+

AI glasses forecast to ship this year. Every pair is a camera we can run on, and we pay $0 to build it.

Sept 2026

Meta opened its glasses' camera to outside apps. The platforms want specialists building on top.

$99 → $25

What expert video help for homeowners fell to, per year. Consumer help is a race to free.

THEY BUILD FOR EVERYONE

Glasses, cameras and one free general assistant that answers questions about a single frame.

WE BUILD FOR ONE TRADE

A coach for maintenance teams: their own procedures, step-by-step state, verified records, a pilot on site.

That's also why we don't start with consumers. Consumer help is their turf and they give it away free. In a maintenance shop someone pays, and the same checklist runs every month.

Omdia AI-glasses forecast · Meta Wearables Device Access Toolkit 1.0 · Frontdoor pricing 2023–24 · Apple glasses reported for 2027

MARKET · US, PER YEAR, BOTTOM-UP

A $306M wedge in a $1.8B first market, in a $9B one

TAM — **$9.2B**

15.4M US workers who follow written procedures, at $50 a month for verified records

SAM — **$1.8B**

2.47M US maintenance and repair techs: coaching new hires + records for everyone

WEDGE — **$306M**

~255k new maintenance techs a year, coached for 6 months at $200 a month

YEAR 3 — **$3M ARR**

1% of new techs: ~2,550 trainees, roughly 100 maintenance teams

BLS OEWS national tables, May 2025 · prices are our hypotheses · year 3 is a goal, not a forecast · math in appendix A3

SAVINGS, PRICE + GO-TO-MARKET

A new tech costs $10–15k to hire and ramp up. We charge $1,200. One saved hour a week pays for it

$10–15k

to hire and ramp up one maintenance tech today

$1,200

per trainee: $200 a month for six months, after a short paid pilot

1 hour

a week of a senior tech's time ($44) is what a $200-a-month seat costs, so saving one hour pays for it

HOW WE GET CUSTOMERS

CUSTOMERS 1–10

I sell them myself
Paid pilots with facilities teams and repair shops, each with a metric agreed up front

CUSTOMERS 10–100

Pilots become proof
Case studies, and Praxis reads procedures straight from the maintenance apps shops already use

CUSTOMERS 100–1,000

Partners sell it
Equipment distributors and glasses makers resell it. One site grows to every site

OUTREACH SO FAR: 14 EMAILS SENT 17 OCT · 56 MORE SCHEDULED FOR 19 OCT

Reliability magazine (hiring + ramp-up) · senior-tech hour: BLS wage ÷ wage share · pricing is our hypothesis (appendix A4)

THE LONG GAME

Every coaching session builds data nobody else has. That data is how we climb from technicians to robots

NOW · 2027

Coach new techs

$0.3M

10 teams · ~250 trainees

NEXT · 2028

Verified records

$1.2M

40 teams · records for every tech

THEN · 2029

Our own models

$3M ARR

100 teams · on-device models, better margins

LATER · 2030+

Robots: two paths

A Train our own robot models on the data we've collected

B Sell the data and our step checker to robot labs

AI LABS ALREADY PAY FOR DATA, AND IT'S GROWING

$2B+ a year

Mercor's run-rate selling expert data to AI labs, doubled in four months

>$100M a year

what robotics companies already spend buying real-world data

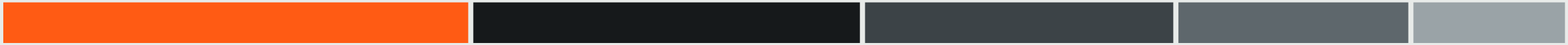
$8.6B by 2030

the AI training-data market, growing about 22% a year

Revenue: targets, not forecasts (~25 trainees a team at $1,200) · Mercor: The Information, Dealroom (Jun 2026) · Micro1 CEO estimate · Grand View Research

THE ASK

$10k buys the data and GPU time to make the models better, and puts Praxis on glasses at a first pilot



$3,000

GPU time

Train our models (step tracker, event detector, local checker) on more data

$2,500

Glasses + capture

Run Praxis on camera glasses at a pilot site

$2,000

Training data

Buy labeled first-person recordings of real maintenance work

$1,500

Inference

About 1,400 hours of live pilot coaching, at ~$1.08 an hour

$1,000

Company formation

So a customer can sign a paid pilot

TRUE BY APRIL

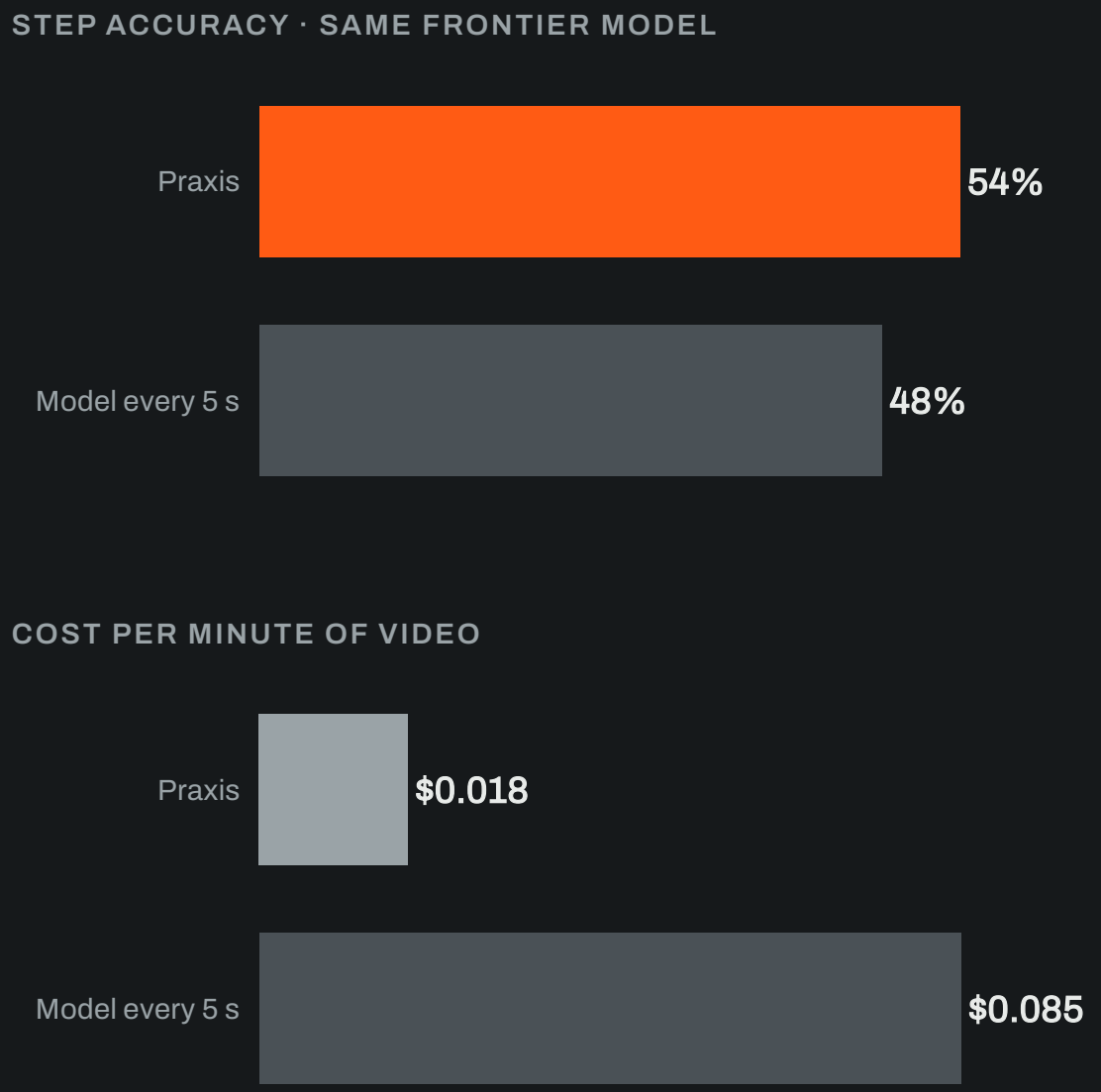
Models retrained on real maintenance recordings

Running on glasses at a real site

A first paid pilot, with a metric agreed up front

PILOT METRIC: TIME UNTIL A NEW TECH WORKS ALONE, AGREED WITH THE CUSTOMER BEFORE WE START

monishsaravana@college.harvard.edu · $10,000 over six months · same amounts as the application



APPENDIX · A1

Same frontier model, with our tracking: 54% vs 48% step accuracy at a fifth of the cost

1/5

the cost per minute of video, and more accurate.

48 held-out public clips · one run each

APPENDIX · A2

The status quo: what each tool does today, and what it misses

SHADOWING

A senior tech, side by side

The real teacher today

MISSES About $44 an hour of senior time, one trainee at a time

MANUALS + VIDEOS

PDFs, binders, YouTube

Explain what should happen

MISSES Can't see what you're actually doing

PROCEDURE APPS

MaintainX, Tulip, Augmentir

Put the steps on a phone; collect sign-offs and photos

MISSES A checkbox isn't proof the step was done right

FIXED CAMERAS

Retrocausal, Invisible AI

Watch one workstation on a line

MISSES Maintenance moves around the machine

PRAXIS

Watches the work, follows the person around the machine, and runs on hardware a shop already owns.

Competitor pages checked 2026-10-17

APPENDIX · A3

Market math, and the assumptions behind it

	WHO (BLS)	MATH	PER YEAR
TAM	Installation, maintenance + repair 6,086,190 · electricians 757,220 · production 8,565,760 = 15.4M	15.4M × $50 a month × 12	≈ $9.2B
SAM	General maintenance + repair, machinery mechanics + millwrights, HVAC = 2.47M jobs, ~255k openings a year	255k × $200 × 6 months = $306M; 2.47M × $50 × 12 = $1.48B	≈ $1.8B
SOM	1% of new maintenance techs ≈ 2,550 trainees a year, ~100 mid-size teams	2,550 × $200 × 6 months	≈ $3M ARR

Prices are hypotheses. $200 a month for coaching anchors on Tulip's $100–250 per station; $50 a month for every-worker records is a guess

Openings include people changing jobs, so new-tech counts are an upper bound

Not counted: other countries, field service outside these groups, aviation maintenance, and the robotics upside

BLS OEWS national tables, May 2025 · BLS Occupational Outlook Handbook

APPENDIX · A4

Unit economics, and the pricing risk

DESIGN	$ / VIDEO-MIN	$ / TRAINEE-MONTH, 2 H/DAY	MARGIN AT $200
Frontier model every 5 s	$0.085	about $214	negative
Praxis engine today	$0.018	about $45	about 77%
Praxis + our local model	$0.008	about $20	about 90%

Labeled assumptions, not sourced: if a senior tech spends 50% of their time on a new hire for 3 months, that's about $11,400 of their time; at 25% for 12 months, about $22,800. We're asking maintenance managers for their real split.

THE RISK

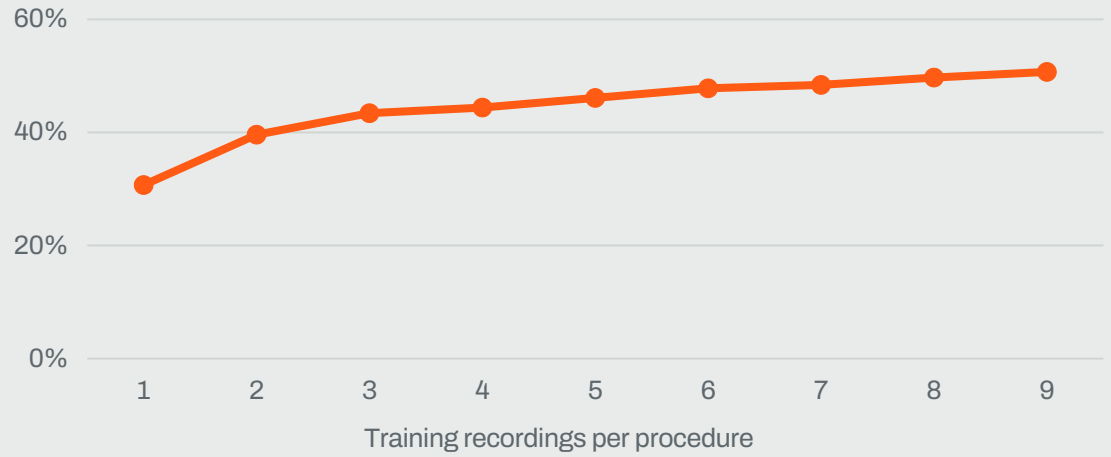
All-shift use (8 h/day) makes a flat $200 seat $1.19 an hour against our $1.08 cost: about a 9% margin. With the local model, about 60%. So price by coached hours or tiers, and move routine checks to the local model first.

2,520 coached minutes a month = 2 h/day × 21 days · senior-tech cost: BLS

Both learning curves: step models and the local student

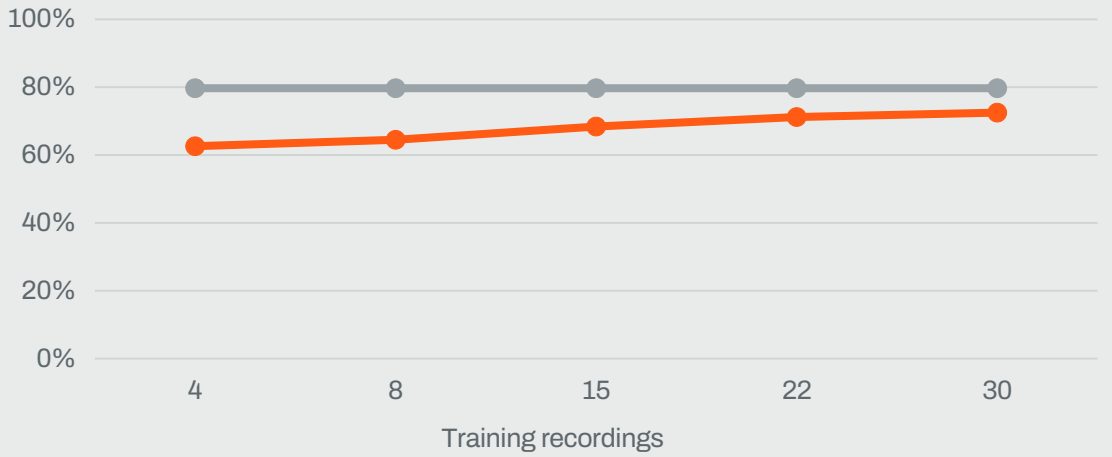
30.7% → 50.7%

recognizing the current step, on videos it never trained on



62.6% → 72.5%

checking a step is done · grey: the frontier model, 79.7%



TESTED ONLY ON RECORDINGS IT NEVER SAW · BOTH STILL RISING AT THE LARGEST SIZE WE HAVE

Random subsets of public train-split recordings, 5 seeds per point

Training data, stated plainly

Train and test never overlap. Everything learned uses only the dataset's official train-split recordings

Small on purpose. A weekend-scale proof: the step-change detector trained on 24 recordings, the step models on 3 per procedure, the student on checks from 30

Tested on 48 held-out clips: 235 min, 302 annotated steps, 170 annotated mistakes, plus procedure types never seen in training

Design choices were made on train-split recordings only, never on the test clips

Labels are the dataset's, not ours. One run per system, no confidence intervals yet

LEARNING CURVES

Step models: 30.7% with 1 recording → 43.4% with 3 → 50.7% with 9 (the 3 procedures with 9)

Student: 62.6% with 4 recordings → 72.5% with 30; the frontier teacher is 79.7%

Detector: noisy, no clear trend yet. Not claimed

APPENDIX · A7

Who else is here, and how we differ

CATEGORY	EXAMPLES	US VS THEM
Instruction platforms	Tulip, Augmentir, Dozuki	They show the steps; we watch the work. Natural integration partners
Line cameras	Retrocausal, Invisible AI, Tulip Vision	Fixed at a station on a production line. Can't follow a person around a machine
First-person coaching	Viture Helix + NVIDIA, Strivr	Closest. Hardware-led or high-touch; no published accuracy
Toolkits	NVIDIA XR AI, Meta's camera toolkit, Android XR	Complements: the plumbing we build on
General assistants	Meta AI, Gemini Live	Answer questions about a frame. No plan, no state, no calibration

If a big platform ships a coach: they're building our hardware and plumbing. We go deep in one vertical, with its procedures, records and data.

Public product pages, checked 17 Oct 2026

Why not consumers first? Because the budget and the repeat data are in business

$3.6B

what Autodesk agreed to pay for MaintainX, an app that puts maintenance procedures on a phone.

$99 → $25

what Frontdoor's expert video help for homeowners fell to, per year, from 2023 to 2024.

The same checklist runs on every machine, every month. Home tasks don't repeat, and the platforms give consumer help away free. Consumers come last.

Autodesk–MaintainX release (28 May 2026) · Frontdoor pricing 2023–2024

APPENDIX · A9

The risks we see, and what we do about each

RISK	WHAT WE DO
Meta or Google ships a coach	They're building our hardware and plumbing. We go deep in maintenance with its procedures, records and data, and run on their glasses
Glasses aren't ready	It works on a tablet or phone today. Glasses are one more input adapter
Mistake detection is weak (4 of 17)	Locked procedures, maintenance data and the tech saying "done" all add evidence. Measured in every pilot
Privacy, unions, monitoring laws	Fast layers run locally; only event keyframes leave (14% of frames). Coaching for the worker, aggregates for the employer. Blur next
A wrong prompt on a safety-critical step	Lockout and similar steps stay the tech's job. Praxis prompts and records them, and never approves them
Depending on model vendors	The verifier is swappable in one config line, and our local student already handles routine checks
Solo founder	The grant tests demand before hiring. First hires come after the pilot

Glasses and voice: what's built, and what's next

BUILT

- Runs live in real time, with the local layers at about 50 ms per frame; model calls never block the loop
- Glasses connections for Meta and Viture + NVIDIA, plug and play
- Voice control, plus screen or voice-only modes depending on the glasses
- Praxis Studio: the engine in a browser, on a camera, an upload or a test clip

NEXT, WITH DEVICES IN HAND

- Test end to end on Meta and Viture glasses at a pilot site
- Measure glass-to-coach latency, the biggest unknown
- A hearing-protection headset for loud plants
- Face and badge blur; a local-model tier for sensitive sites

Built; not yet tested on the devices

I've built real-time vision on messy video before, and I built this in a week

LipTrac

Real-time lip reading from ordinary video. 60,000+ frames, 12 testers, provisional patent filed.

7 years

of competitive robotics, coaching teammates without seeing what they see.

1 week

to build the engine, a 48-video eval and a live app, solo. First commit Oct 17.

Monish Saravana · Harvard College · solo founder. Also: path planning for oil-cleanup boats from aerial imagery.